

StickForStats: A Statistical Analysis Platform with Automatic Assumption Validation

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Summary

StickForStats is an open-source, web-based statistical analysis platform that automatically validates the assumptions underlying every statistical test before execution. Its core innovation is the Guardian system—a pipeline of eight validators that intercept each analysis request, check assumptions such as normality, variance homogeneity, independence, and outlier presence, and either proceed with full confidence reporting or automatically cascade to an appropriate nonparametric alternative. The platform pairs this validation layer with a natural-language interface that lets researchers describe their question in plain English, an autonomous test-selection engine, plain-language result interpretation, and a manuscript review module that extracts and re-checks statistical claims from uploaded papers. StickForStats is built with a Django REST backend serving 195 API endpoints, a React frontend with 25 pages supporting 16 languages, and optional Celery-based asynchronous processing for long-running analyses.

Statement of Need

Statistical assumption violations are pervasive in published research. Surveys across psychology, education, and biomedicine consistently find that researchers rarely verify the assumptions of the tests they apply (Hoekstra et al., 2012; Keselman et al., 1998). When assumptions such as normality or homogeneity of variance are violated, Type I and Type II error rates can deviate substantially from their nominal levels (Zimmerman, 2004), undermining the validity of conclusions. The consequences compound at scale: the broader reproducibility crisis in science has been attributed in part to widespread statistical misapplication (Baker, 2016; Ioannidis, 2005; Open Science Collaboration, 2015).

Existing statistical software places the burden of assumption checking entirely on the analyst. General-purpose environments such as R (R Core Team, 2023) and Python's SciPy (Virtanen & others, 2020) require users to explicitly call separate diagnostic tests and interpret the results themselves. GUI-based tools like JASP (JASP Team, 2023) and jamovi (The jamovi project, 2023) simplify test execution but still leave assumption verification as an optional, manual step. No widely adopted platform currently intercepts the analysis pipeline to enforce assumption checking automatically and transparently.

StickForStats addresses this gap. Its target users are researchers in the social, behavioral, and biomedical sciences who regularly apply parametric statistical tests but may lack the expertise or time to conduct comprehensive assumption diagnostics. The platform is equally useful for statistics instructors who want students to learn correct analytical workflows, and for journal editors and reviewers who need to verify reported statistics in manuscripts.

The Guardian System

The Guardian is a middleware layer that wraps every statistical test endpoint. When an analysis request arrives, the Guardian selects the relevant subset of its eight validators and runs them against the submitted data before the primary test executes.

The eight validators and their methods are:

1. **Normality** — Shapiro-Wilk (Shapiro & Wilk, 1965) and Anderson-Darling (Anderson & Darling, 1954) tests on each group.
2. **Variance homogeneity** — Levene's test (Levene, 1960) with Brown-Forsythe correction (Brown & Forsythe, 1974).
3. **Independence** — Autocorrelation analysis (lag-1 through lag-5) with Durbin-Watson statistic (Durbin & Watson, 1951).
4. **Outlier detection** — Combined IQR fencing and modified Z-score method (Grubbs, 1969).
5. **Sample size adequacy** — Rule-based minimum thresholds calibrated per test type, informed by power analysis literature (Cohen, 1988).
6. **Modality** — Kernel density estimation with bandwidth selection to detect multimodal distributions.
7. **Linearity** — Comparison of linear and polynomial R^2 values supplemented by the Wald-Wolfowitz runs test (Wald & Wolfowitz, 1940).
8. **Homoscedasticity** — Breusch-Pagan test (Breusch & Pagan, 1979) for regression residuals.

Each validator returns a severity level (critical, warning, or minor) and a weight ($w = 3.0, 2.0, 1.0$ respectively). The Guardian computes a composite confidence score:

$$C = \max\left(0, 1 - \frac{\sum w_i}{W_{\max} \times 1.2}\right)$$

where $W_{\max}$ is the maximum possible penalty for the test type. A score above 0.8 indicates the analysis can proceed with confidence; between 0.6 and 0.8 signals caution; below 0.6 triggers review. When any critical violation is detected, the AutonomousCascadeEngine automatically re-routes the analysis to the most appropriate nonparametric alternative (e.g., Mann-Whitney U for a failed independent t -test) with full documentation of the decision path.

The Guardian system is validated by 38 automated tests (22 integration, 16 middleware) that run in continuous integration on every commit.

Additional Capabilities

Autonomous analysis pipeline. The SmartProfiler module automatically detects variable types, distributions, and data quality issues from uploaded datasets. Combined with intent detection from natural-language queries, the platform selects and executes the appropriate test without requiring the user to specify it. Results are translated into plain-language summaries via a template-based PlainLanguageTranslator.

Manuscript statistical review. StickForStats can parse PDF, LaTeX, and DOCX manuscripts, extract statistical claims using a regex and LLM hybrid pipeline, and re-check each claim for internal consistency in the style of STATCHECK (Bakker & Wicherts, 2011). Seven specialized validators assess completeness, consistency, power reporting, multiple-comparison corrections, assumption documentation, effect-size reporting, and reproducibility. Discipline-aware profiles (medicine, psychology, economics) weight validators according to field-specific reporting standards such as CONSORT (Schulz et al., 2010) and JARS-Quant (Appelbaum et al., 2018).

83 **Comprehensive statistical coverage.** The platform implements parametric and nonparametric
84 tests, correlation and regression (including ridge and lasso), categorical tests, meta-analysis with
85 DerSimonian-Laird random effects (DerSimonian & Laird, 1986) and Egger's publication bias
86 test (Egger et al., 1997), power analysis validated against G*Power (Faul et al., 2007), survival
87 analysis, factor analysis, causal inference (DAGs, propensity scores, mediation, difference-in-
88 differences), mixed models, and missing-data handling.

89 **High-precision computing.** An optional 50-decimal-digit precision mode using mpmath
90 (Johansson, 2013) is available for all core calculations, critical for validation studies where
91 floating-point accumulation matters.

92 **Statistical Quality Score.** Forty-five rules across six categories (test selection, assumption
93 reporting, effect sizes, confidence intervals, multiple comparisons, reproducibility) produce an
94 automated quality score (0–100) for any analysis or manuscript.

95 Validation

96 All statistical calculations are validated against SciPy (Virtanen & others, 2020) and R (R
97 Core Team, 2023) reference implementations. Parametric tests agree to 14–16 decimal places;
98 meta-analysis results agree to 10 decimal places; power analysis results agree with G*Power
99 within 1%. Reproducibility scripts and reference datasets (Fisher's Iris (Fisher, 1936), UCI
100 Wine Quality (Cortez et al., 2009)) are provided in the repository under paper/replication/.

101 AI Disclosure

102 Development of StickForStats was assisted by Claude (Anthropic). All AI-generated code
103 was reviewed, tested against reference implementations, and validated through the project's
104 continuous integration pipeline (515 backend tests, 573 frontend tests, all passing). Statistical
105 correctness was verified independently against SciPy, R, and G*Power.

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